

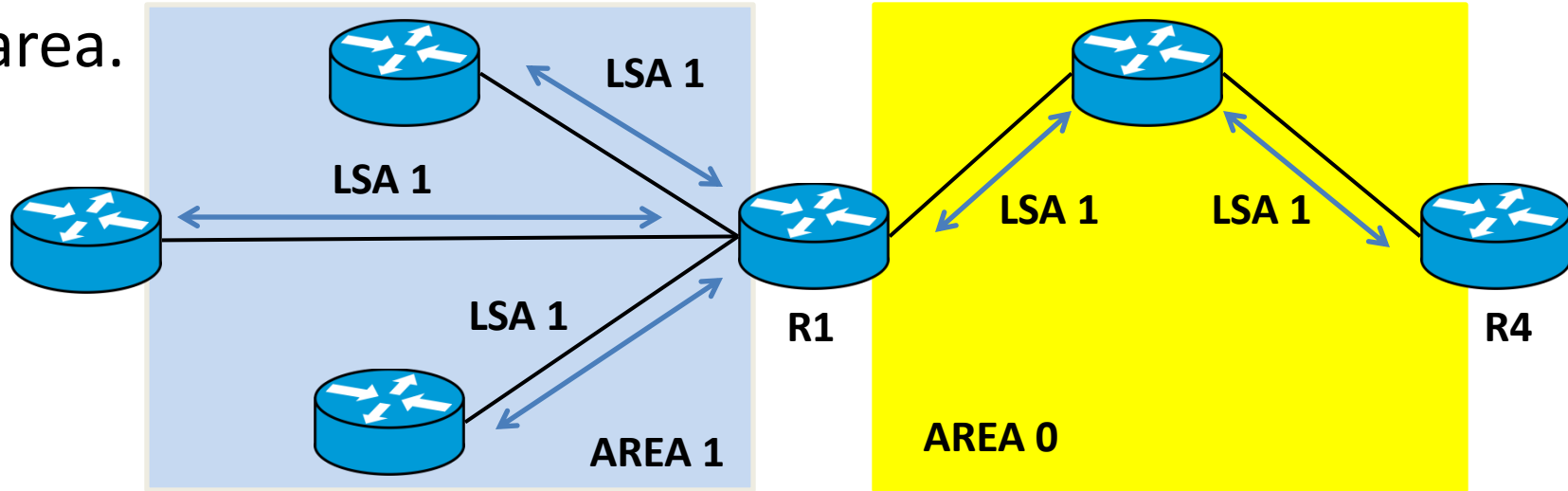
LSA Types:

LSA" stands for Link-State Advertisement, a crucial part of the OSPF (Open Shortest Path First) routing protocol. LSAs are used to communicate the state of routers and network segments to other routers within an OSPF area.

- 1. Type 1 - Router LSA**
- 2. Type 2 - Network LSA**
- 3. Type 3 - Summary LSA**
- 4. Type 4 - ASBR Summary LSA**
- 5. Type 5 - External LSA:**
- 6. Type 6 - Multicast LSA (Note: Rarely used)**
- 7. Type 7 - NSSA LSA:**
- 8. Type 8 - External Attributes LSA:**
- 9. Type 9, 10, and 11 - Opaque LSAs:**

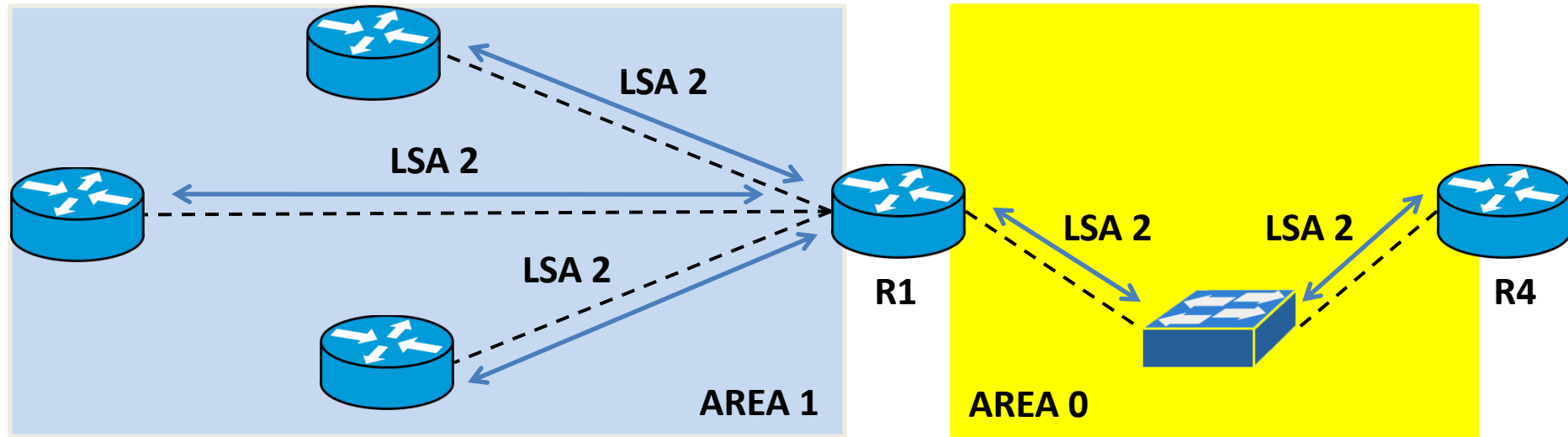
Type 1: Router LSA: -

- Generated by every router.
- Contains information about the router's directly connected OSPF-enabled interfaces and their states, as well as the router's link cost.
- Flooded within an area, meaning it doesn't go beyond the OSPF area.



Type 2: Network LSA:

- Generated by the **Designated Router (DR)** in a broadcast or non-broadcast multi-access network.
- Represents the multi-access network and lists all routers that are attached to the network.
- Flooded within an area.

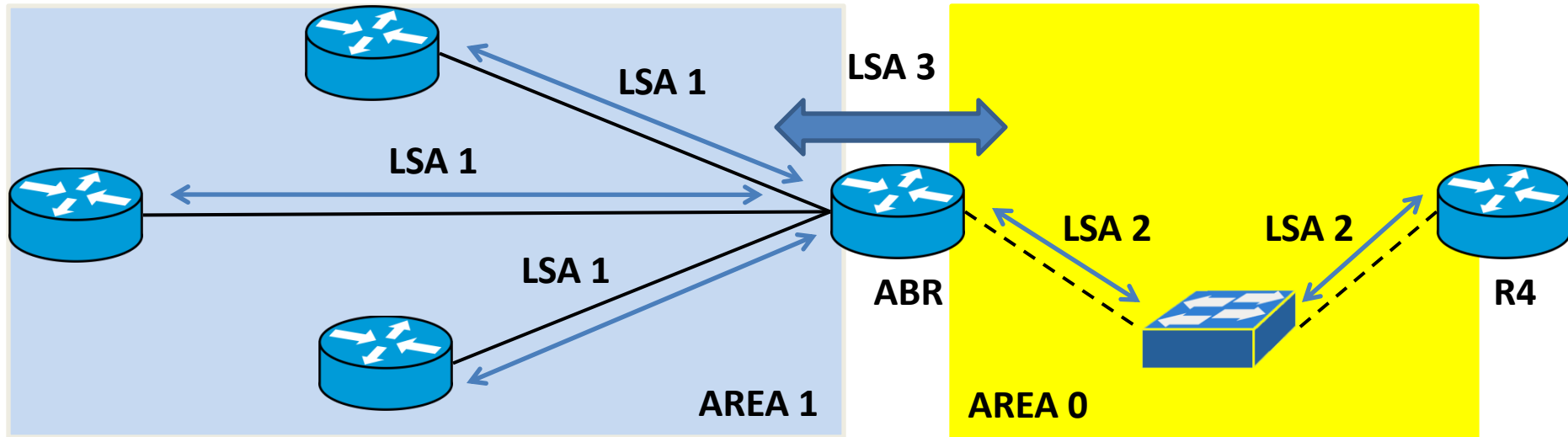


Type 3: Network LSA: -

Generated by Area Border Routers (ABRs).

Advertises network information from one area into another (i.e., summaries of Type 1 and Type 2 LSAs from one area to another).

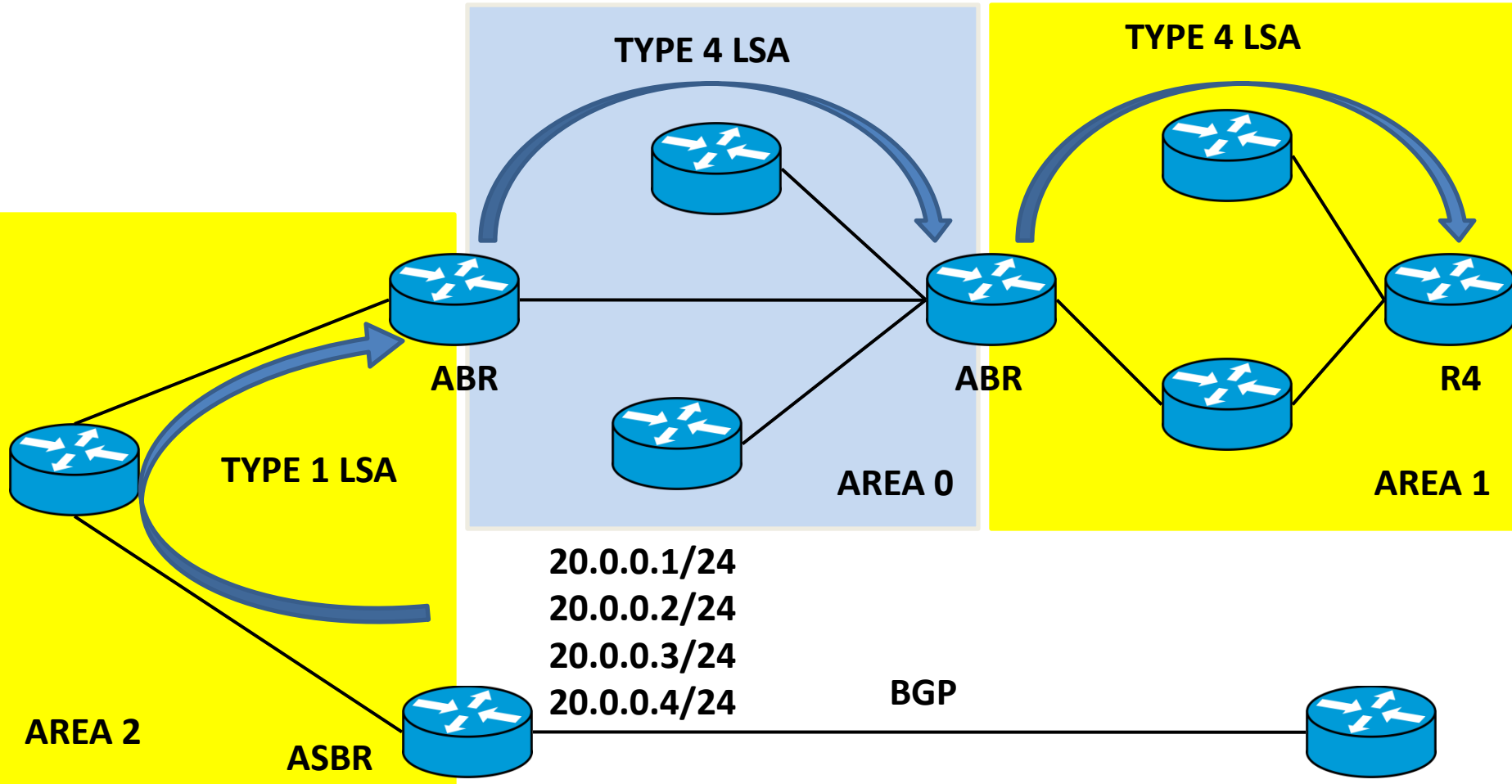
Flooded into the entire OSPF domain.



Type 4: ASBR Summary: -

- Generated by ABRs.
- Advertises the route to an **Autonomous System Boundary Router (ASBR)** to other areas in the OSPF domain.
- Provides reachability information to external routes.

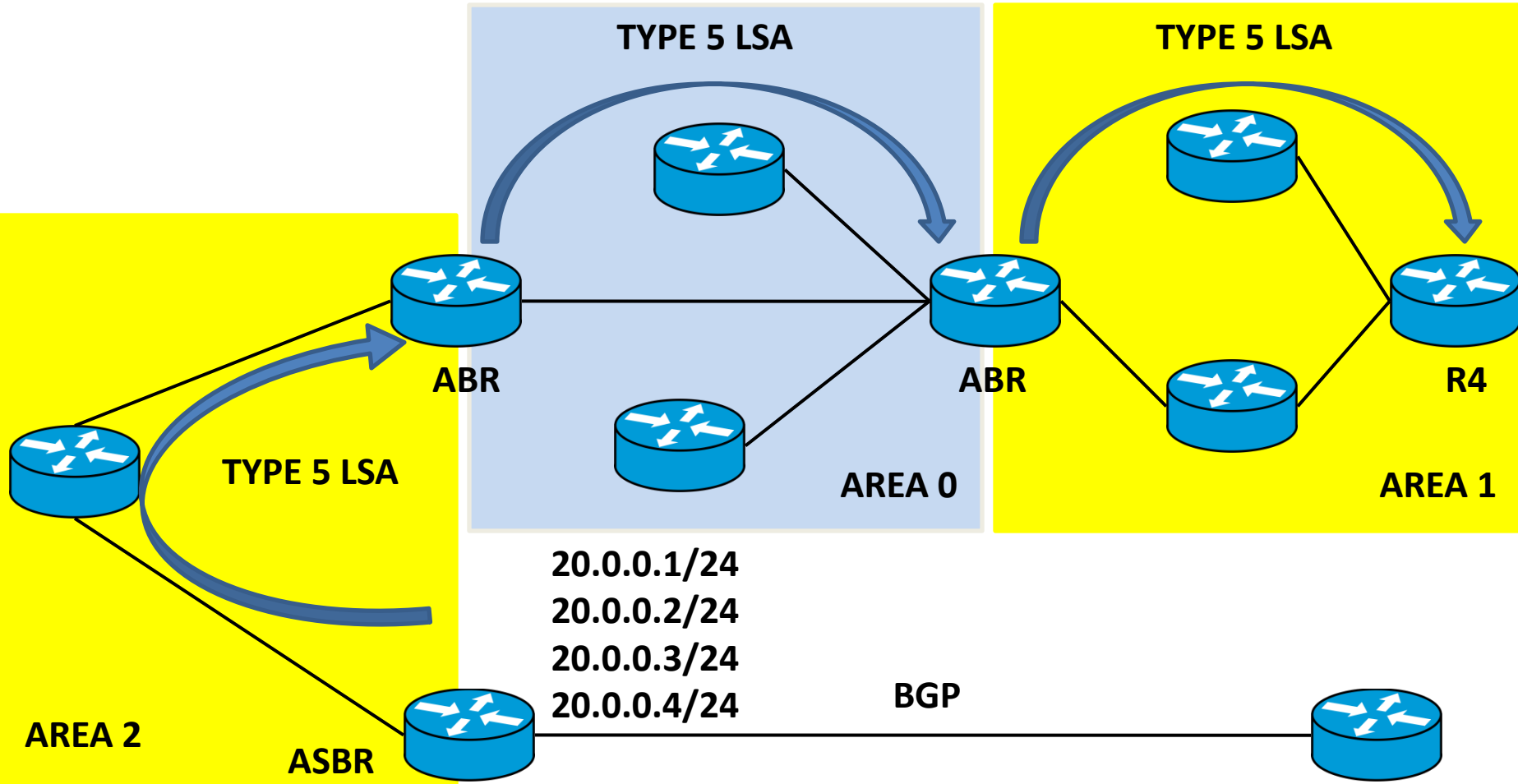
Type 4: ASBR Summary: -



Type 5: AS External LSA: -

- Generated by ASBRs.
- Advertises routes to external networks (networks outside the OSPF domain).
- Flooded throughout the entire OSPF domain, except for stub areas.

Type 5: AS External LSA: -



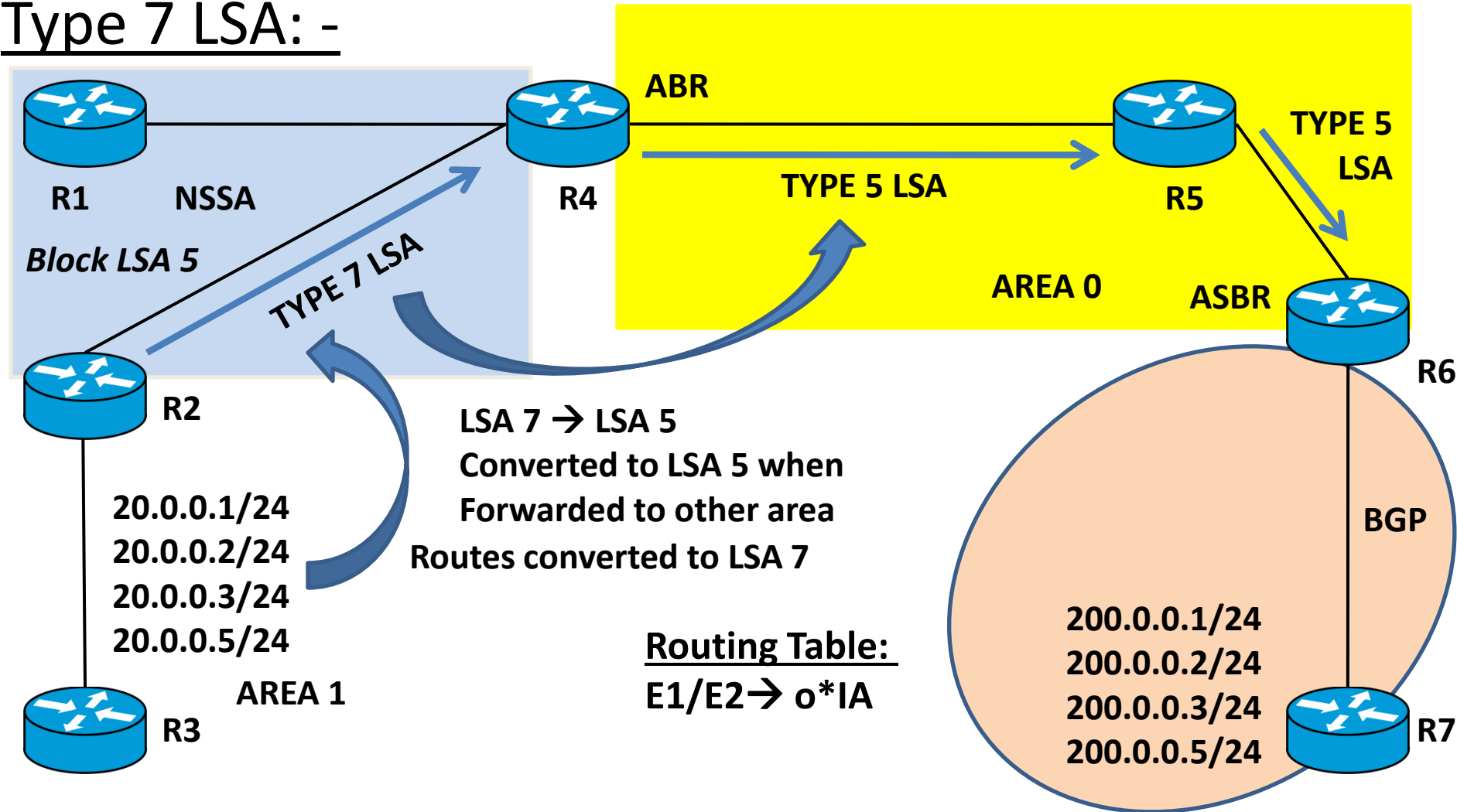
Type 6: Multicast OSPF LSA (MOSPF)

- Used for multicast routing.
- Not commonly implemented and largely deprecated in modern networks.

Type 7: NSSA LSA: -

- Generated by ASBRs within a **Not-So-Stubby Area (NSSA)**.
- Translates external routes into the OSPF domain for NSSAs.
- Can be converted into a Type 5 LSA by an ABR.

Type 7 LSA: -



Type 8: External Attributes LSA (used in BGP)

- Used in **OSPF for Border Gateway Protocol (BGP)** to carry additional routing information.
- Less common in typical OSPF implementations.

Type 9-11: Opaque LSAs

- Used for future extensions and enhancements.
- Type 9 is scoped to the local link, Type 10 is scoped to the area, and Type 11 is scoped to the entire OSPF domain.
- Commonly used for **Traffic Engineering (TE)** information.